Can Recent Surveillance and Weather Data Flag Dengue Surge Risk?

Singapore, weekly forecasts

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Statistical Learning

December 2026

Decision question

For Singapore dengue surveillance teams

Can information available by week t help identify a materially elevated dengue fever (DF) week at $t + h$?

- Forecast horizons: 1, 2 and 4 weeks
- Compare recent cases, weather, and their combination
- Predictive association only; this is not a causal estimate or live alert

Why is an early warning difficult?

- Dengue is endemic, but incidence and outbreak timing vary substantially.
- Recent counts are powerful, but can make a model look good through persistence.
- Weather may add useful lead time, but the weather sources disagree.
- A false alarm consumes response capacity; a missed surge delays preparation.

Key distinction

Predicting that an ongoing high period continues is not the same as warning before onset.

Data: national cases plus local weather

- MOH weekly bulletin, 2012–2022 (574 epi weeks)
- Target: Dengue Fever; DHF kept separate (rare)
- NEA/MSS Changi (S24): rainfall and temperature
- Open-Meteo: humidity; checked against NEA monthly RH
- MOH epi weeks mapped to the following Monday

- 0 duplicated or missing case weeks
- MOH releases identical for 209/261 DF weeks
- Max revision: 8 cases/week; totals differ by 0.2%
- Weather agreement: temperature $r = 0.91$, rain $r = 0.48$

Three meanings of a surge

Let y_t be weekly DF cases and h the horizon:

1. **Threshold A:** $y_{t+h} > \text{mean}_{52,t} + 2\text{SD}_{52,t}$
2. **Seasonal S:** above the historical same-season 90th percentile
3. **Growth G:** at least 50 cases and a 25/30/45% rise for $h = 1/2/4$

Onset A/S only count weeks when the current series is not already above threshold. At $h = 1$, A–S Jaccard overlap is 0.31; S–G is 0.05.

What goes into the models?

- Case history: log counts, lags, rolling means, recent growth
- Season: week-of-year and Fourier terms
- Weather: rain, rain days, temperature and humidity at lags 0–12 weeks

Model families

Prior, seasonal and persistence baselines; logistic regression (cases/weather/full); lasso logistic; CART; random forest; Poisson GLM; ridge on log counts.

Count forecasts are converted to probabilities of crossing each surge threshold.

Validation: respect time and label availability

- Expanding-window rolling origin by week; evaluation begins in 2016.
- Refit weekly; purge examples whose full label horizon is not yet known.
- Development outcomes are before 2022; final-period outcomes are in 2022.
- Earlier observed 2022 labels can enter training for later rolling forecasts.
- Metrics: PR-AUC, precision, recall, Brier/calibration, alarms/year; count MAE/RMSE in cases/week.
- 95% moving-block bootstrap intervals (26-week blocks).

Important qualification

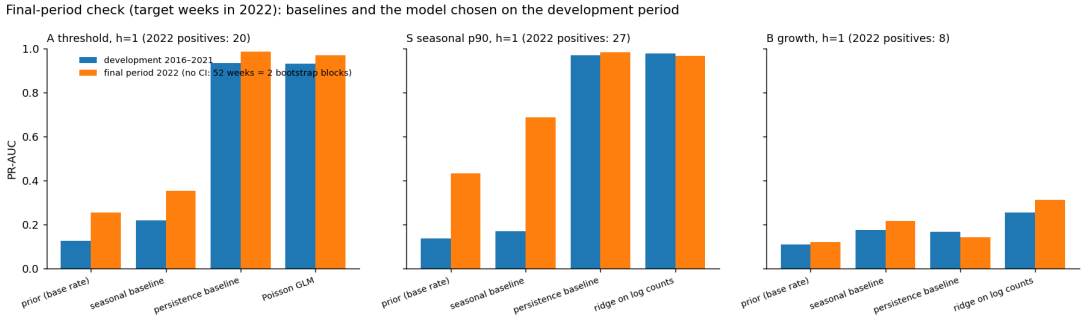
Earlier exploratory runs included forecasts through 2022. This is not a pristine static test set; describe it as a final-period prequential check.

Development results ($h = 1$, target weeks 2016–2021)

Target	Pos.	Persistence	Best model: PR-AUC [95% CI]	P	R
A threshold	43	0.93	Poisson GLM 0.93 [0.78, 0.98]	0.62	0.93
S seasonal	44	0.97	Ridge log-count 0.98 [0.77, 1.00]	0.75	1.00
G growth	36	0.17	Ridge log-count 0.26 [0.15, 0.40]	0.22	0.53
A-onset	7	0.30	Poisson GLM 0.32 [0.04, 0.80]	0.17	0.57
S-onset	5	0.26	Ridge log-count 0.89 [0.70, 1.00]	0.33	1.00

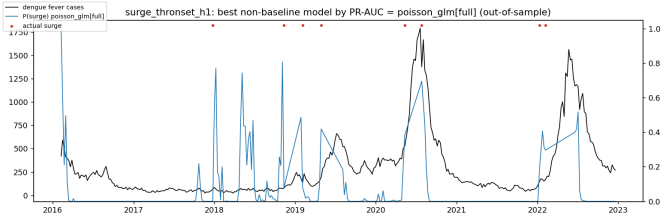
- Level surges (A, S): persistence is as good as any model, so this is not early warning.
- Growth and S-onset: count regression beats the baselines, but S-onset has only 5 events.
- Weather-only logistic: 0.34 / 0.16 / 0.13, close to the base rates; adding weather does not help.

2022 final-period check



Ranking holds: A/S persistence 0.98 vs selected model 0.97; growth 0.31 vs persistence 0.14 (8 positives). Onset targets have only 2 positives in 2022, too few to compare models.

Failure case: alarm at the 2022 peak



- Target: A-onset, $h = 1$; model: Poisson GLM
- Origin 2022-07-04: forecast 1,383 cases, $P = 0.52$, so an alarm is raised
- Observed: 918 cases. The epidemic had peaked
- The model extrapolated recent growth into a turning point
- It also missed the 2022-01-17 onset ($P = 0.18$, just below the 0.20 threshold)

Limitations and responsible use

- Few outbreaks and very few onset positives mean wide uncertainty.
- No serotype, immunity, spatial exposure or local case counts.
- Notified cases can shift with testing and reporting; COVID-19 is a special period.
- One weather station and weaker rainfall/humidity agreement limit input quality.
- The assumed 4:1 cost ratio is illustrative, not elicited from decision-makers.

Use

Retrospective research comparison only; never use these scores as deployed public-health alerts without prospective validation and human review.

Takeaways

1. Surge definitions materially change the warning task.
2. Persistence must be beaten before claiming useful early warning.
3. Count regression helps for growth and onset surges, but few events mean wide intervals.
4. Weather adds no measurable skill beyond recent counts at 1–4 weeks.

Questions?

Reproducibility and team contributions

- Pipeline, configuration, processed weekly snapshots and source manifest are included.
- Main run: `cd pipeline && .venv/bin/python run.py all`
- AI-use disclosure: `AI_USE.md`
- Contributions: [to be completed by the group]